Startup Guide

FANUC CR-15*iA*

Thank you for purchasing a FANUC Robot

READ THIS FIRST

Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation provided with your robot for safety procedures.

Required Tools

- Straight-head screwdriver
- Cross-head screwdriver
- Torque wrench
- Metric hex-head wrenches

Required Items

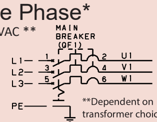
- Properly rated transport equipment as specified
- Customer-supplied properly rated power source as specified on the Power Requirements Label
- Multimeter
- Loctite #243
- Rigid platform to hold robot
- Torque wrench (270Nm)
- 4-M16 bolts
- 4-M16 flat washers

Optional Items

- Robot locator pins
- Auto Confirmation Signal switch

Three Phase*

440-500VAC **



**Dependent on transformer choice.

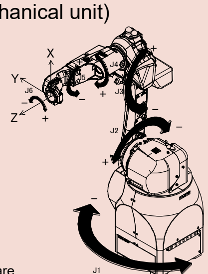
*When using a Mate controller, Single Phase is an available option.

R-30*i*B Plus Controller*

Controller

*Although not shown, the Mate controller is also available.

Robot (mechanical unit)



Note: all axes are 0° in this position.

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A. Unpack and Install your Robot

START HERE

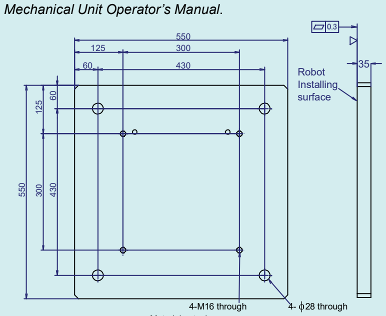
Shipment Contents

Your shipment will arrive on two skids. One will contain the Robot Mechanical Unit. The other contains the Controller and Accessory Boxes.

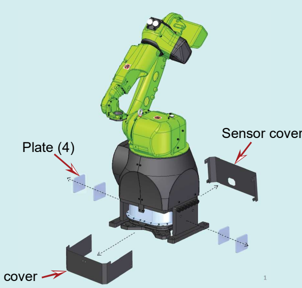
1. Remove the bolts holding the top of the container and remove the top. Carefully loosen the bolts on each side and remove one side at a time. Unwrap and properly dispose of the plastic around the robot.

2. Remove the bolts holding the robot to the skid.

3. Prepare an installation plate as shown below. Make sure that the area to which the robot will be mounted is not painted or fabricated to ensure that the area is level. More information is available in the *CR-15iA Mechanical Unit Operator's Manual*.

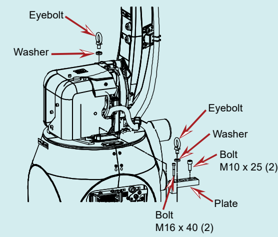


4. Remove the plates and the sensor covers.



5. Hang the robot with a crane.

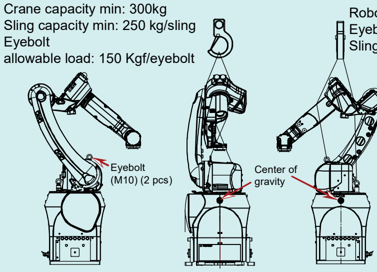
Installing transport equipment



Transportation using a crane

Crane capacity min: 300kg
Sling capacity min: 250 kg/sling
Eyebolt
allowable load: 150 Kgf/eyebolt

Robot mass: 255kg
Eyebold qty: 2
Sling qty: 2

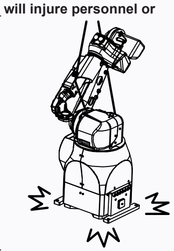


Robot transportation posture
J1: 0°
J2: -30°
J3: -40°
J4: 0°
J5: -50°
J6: 0°

⚠ WARNING

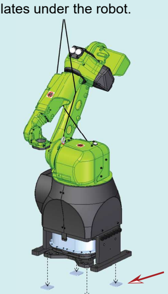
Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

When hoisting or lowering the robot with a crane, move it slowly and with great care. When placing the robot on the floor, exercise care to prevent the installation surface of the robot from striking the floor with too much force.

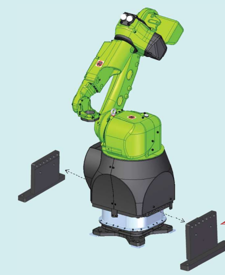


When hoisting or lowering the robot, be sure to perform it with the robot only. Do not hoist or lower the robot with a pedestal, or an installation plate and skid for transportation.

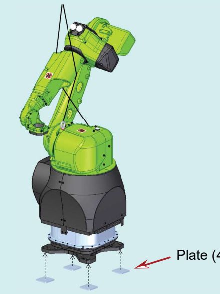
6. insert the 4 plates under the robot.



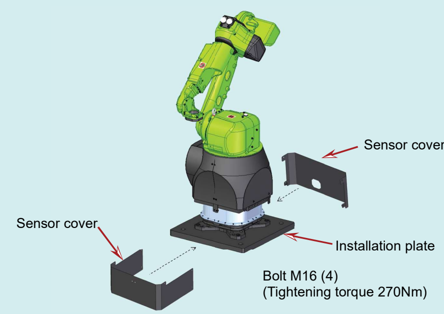
7. Lower the robot to the ground, then remove the transport equipment.



8. Hang the robot with a crane, then transport it to the installation place.



9. Place the robot on the installation plate, tighten bolts then attach the sensor cover.



NOTE: Do not dispose of the transport equipment after installation.

Robot installation is complete. You are now ready to go to **B. Unpack and Install your Controller**.

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Startup Guide

FANUC CR-15*iA*

B. Unpack and Install your Controller

1. Open each of the boxes that came with your robot. One will contain the items below.

Hardware Bag, Sensor Calibration CD, Teach Pendant, & Inspection Data Sheet

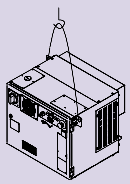
Store these items in a safe, known location.

2. Use properly rated equipment to remove the controller from the skid.

Minimum crane capacity: 300kg
Minimum sling capacity: 300kg

⚠ WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.



3. Place the controller in a safe location where the grounding wire, RMP cable, and sensor cable can reach the robot, and the controller is safely reachable from outside of the robot work area.

1 - Sensor cable (square connector)
2 - Teach Pendant cable (round connector)
3 - RMP cable (rectangular connector) **NOTE:** this cable is also called the Robot Connection Cable (RCC)
4 - Ground wire

4. Attach the RMP cable from the robot to the controller.

5. Attach the ground wire from the controller to the robot.

6. Attach the sensor cable from the controller to the robot.

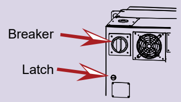
7. Determine the power requirements for your controller from the label on the front. The label is at the bottom left corner of the controller door.

NOTE: The following steps refer specifically to the R-30iB Plus controller. The concepts, however, apply to the Mate controller as well. For details regarding a Mate controller, Refer to *Connecting the Input Power* found in the FANUC *Robot series R-30iB mate/R-30iB Mate Plus CONTROLLER MAINTENANCE MANUAL* .

⚠ WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

8. Open the controller door by using a straight-head screwdriver to turn the latch clockwise, then turn the Breaker counter-clockwise past its off position.

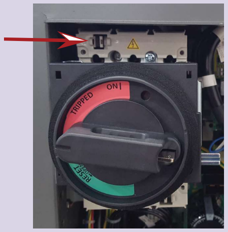


9. Unscrew the round white plug from the side of the controller. Remove the four screws highlighted in the illustration below, and carefully remove the Breaker access panel. It may stick as it has a sealant around the edges.

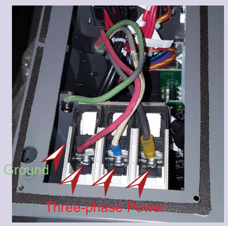
10. Route the user-supplied input power cable through the side hole. It is typical to purchase and install a strain relief cable grip (example shown below) to remove stress on the power cable connections and isolate the power cable from the controller cabinet.

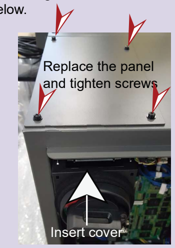
11. To connect power, remove the plastic cover from above the Breaker (see the diagram below.) To remove the cover, insert a straight-head screwdriver into the opening, and push the tab to the left allowing the cover to slide forward.

Insert a straight-head screwdriver and push the tab left. Use your other hand to pull the cover forward.



12. Verify that the input power cable is not energized. Connect it to the terminals as shown on the left below.

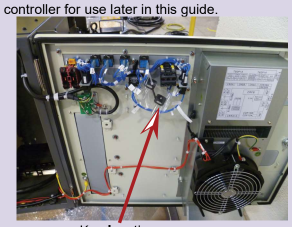




13. Re-insert the Breaker cover and click it into place. Place the Breaker access panel back over top of the breaker and secure with four screws.

14. With the breaker still off, energize the input power cable and use a multimeter to verify the correct voltage is present via the holes in the breaker cover.

15. Before you close the controller door, cut loose the set of keys from inside the door, and store them in a known location outside of the controller for use later in this guide.



Key location inside controller

16. Close the controller door and use a straight-head screwdriver to turn the Latch on the outside of the controller counter-clockwise.

Controller installation is complete. You are now ready to go to **C Turn on Your Controller**.

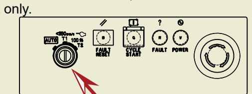
C. Turn on your Controller

⚠ WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

1. Turn on the controller by rotating the Breaker clockwise to ON.

2. Locate the Mode switch on the Controller Operator Panel. Use the key retrieved in Step B15 to switch to T1 mode. T1 mode limits the robot speed to 250 mm/sec and requires robot control to be at the teach pendant only.

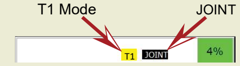


Mode Switch

3. Verify the controller is in T1 Mode by looking at the top right corner of the teach pendant screen. The mode indicator will be hilighted yellow and look similar to the status bar below. If it is not T1, repeat the previous step.

4. Press the COORD key on the teach pendant until JOINT is displayed similar to the status bar below.

T1 Mode JOINT



Teach Pendant Status Bar

You have turned on your controller. You are now ready to go to **D. Jog your Robot**.

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D. Jog your robot

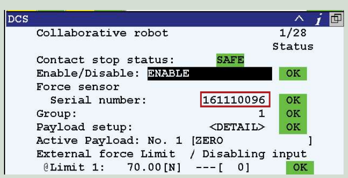
NOTE: If you have loaded new software or if the serial nmber on the DCS Collaboartive screen does not match the serial number on the sensor, perform steps 1-7. **If not, proceed to step 8.**

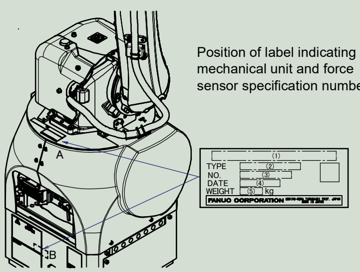
1. Using a computer, load the .cm file from the Sensor Calibration CD (see Step 1) to a USB disk.

2. Insert the USB into the robot controller and navigate to where the .cm file was saved. (MENU>>FILE>>File. F5, [UTIL]>>Set Device>>USB Disk (UD1:)).

3. With the .cm file highlighted, press ENTER to execute the .cm file. You will be asked if you want to execute the file, press F4, Yes.

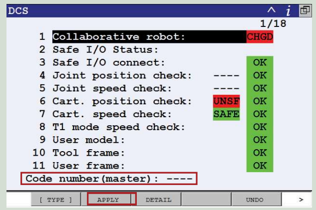
4. Once the file is loaded, the sensor serial number should be shown in the DCS Collaborative robot screen. This number should match the collaborative sensor serial number that is physically on the robot.



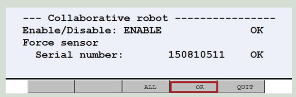


Position of label indicating mechanical unit and force sensor specification number

5. Press MENU, choose NEXT, cursor to SYSTEM > DCS, and press ENTER. Press F2, APPLY, you will be prompted for the DCS code number. Enter the default of 1111.

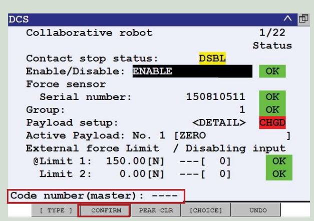


6. On the following screen, press F4, OK, to continue.

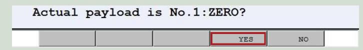


7. You must cycle power on the controller. Turn the Breaker off, wait 5 seconds, and then turn it back on. Alternatively, you can press the FCTN key, cursor to 0 NEXT, and press ENTER. Then cursor to 8 CYCLE POWER, press ENTER, and select YES.

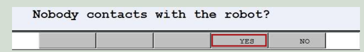
8. After the robot has cycled power, the screen below will be displayed. Press F2, CONFIRM, you will be prompetd for the DCS code number. Enter the default of 1111.



9. You will be prompted with the current payload schedule. If No. 1 is displayed similar to below, confirm by pressing F4, YES. If another number is displayed press F5, NO, and repeat the procedures starting at Step D1.



10. You will be prompted to check that no person or object is in contact with the robot arm. When verified, press F4, YES.



You should see "Payload confirmation success" at the bottom of the screen. Continue to the next step.

11. Ensure the Emergency Stop buttons on the teach pendant and controller are popped out. If not, twist the button clockwise to reset.

12. Press and hold a DEADMAN switch on the back of the teach pendant to its middle position. Press RESET on the teach pendant to clear alarms.

NOTE: To view alarms, press MENU, cursor to ALARM, and press ENTER. To toggle betweeen alarm history and active alarms, press F3.

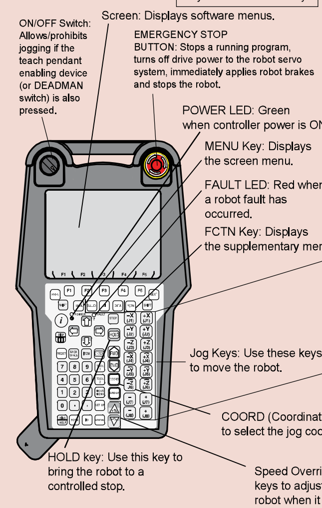
13. To move each joint axis of the robot, press and hold SHIFT and press each jog key one at a time and watch the robot joint move. Refer to the **Robot (mechanical unit) with joints indicated** diagram on Page 1 of this guide.

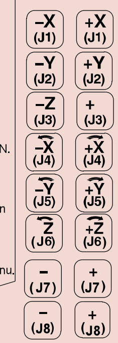
Try jogging each joint of the robot.

Turning on and Jogging the robot is complete.
You are now ready to perform software installation, if necessary.
Refer to the *Controller-specific Software Installation* manual.

Reference

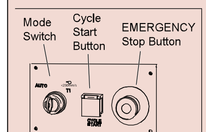
Your actual teach pendant key locations could vary.



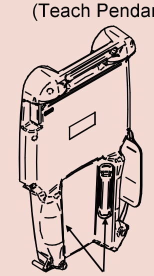


Refer to your *Application-Specific Setup and Operations Manual* for more information on each teach pendant key.

*i*Pendant
(Teach Pendant)



Controller Operator Panel



DEADMAN switch

For more information

To access controller, robot, or application-specific manuals, scan the **QR Code** to the right or go to **https://www.fanucamerica.com/**.

United States: 1-888-326-8287
Canada: 1-905-812-2300
Mexico: + 52 (449) 922-8000
South America: + 55 (11) 3619-0599

Operating Space and Dimension